

The Spatial Diffusion of Labor Strikes*

Evidence from Global Subnational Event Data

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ABSTRACT

Theoretical accounts of labor mobilization emphasize demonstration effects and geographic diffusion, yet quantitative research rarely employs identification strategies that distinguish genuine contagion from common shocks. I examine the spatial structure of strike diffusion using a validated subnational panel of first-level administrative units observed weekly from 2015 to 2025, constructed from GDELT event records geocoded to ADM1 boundaries. Two-way fixed effects models with unit and country-by-week fixed effects absorb both time-invariant confounders and national-level shocks. I find that regions with a striking neighbor are 3.5 times more likely to strike in the raw data, but this association fully disappears once national dynamics are controlled. Decomposing geographic and national channels explicitly reveals large national contagion effects, while geographic neighbor exposure adds nothing once country-level activity is controlled. Moderation analyses further show that freedom of expression weakens the geographic concentration of strike waves, while freedom of association strengthens it, with no corresponding variation in national contagion. These findings suggest that strike contagion operates primarily at the national level, while institutions shape the spatial form of national strike waves.

Keywords—labor strikes, strike diffusion, spatial contagion, subnational panel data, GDELT, collective action

*Early draft. Please do not circulate or cite.

Introduction

Strike waves are among the most extensively documented patterns in the comparative study of labor. Labor historians have traced such waves in Western European industrial societies across more than a century of conflict ([Shorter 1974](#); [Cronin 1987](#); [Franzosi 1989](#)), economists have modeled the business cycles, wage dynamics, and common macroeconomic shocks that produce synchronized strike timing ([Ashenfelter and Johnson 1969](#); [Hibbs 1976](#); [Kennan and Wilson 1993](#)), and scholars of contentious politics have documented comparable cycles across post-communist societies, democratic transitions, and austerity episodes in the developing world ([Beissinger 2002](#); [S. G. Tarrow 2011](#)). Across these traditions and contexts, strike activity has been consistently shown to concentrate into broadly synchronized episodes, but what drives this synchronization remains unclear.

A central question is whether strikes follow a pattern of contagion, whereby mobilization in one region directly raises the probability of strikes in neighboring regions, or whether spatial clustering reflects a national dynamic, with shared economic conditions, union federations, or government policy simultaneously activating workers across all regions at once. The two mechanisms are observationally equivalent, as both produce the spatial clustering documented in the historical literature and visible in contemporary data. Existing scholarship has long leaned toward the national account. Shorter and Tilly ([1974](#)), Cronin ([1987](#)), and Franzosi ([1989, 1995](#)) all conclude that the major strike waves were nationally synchronized, driven by macro-level political and economic cycles rather than sequential geographic spread, and economic models of strike activity point to the same conclusion ([Ashenfelter and Johnson 1969](#); [Hibbs 1976](#)).

Causal identification of diffusion mechanisms has remained elusive, however. Studies of protest diffusion find that geographic neighbor effects attenuate substantially once shared political opportunity structures are controlled ([Kristian S. Gleditsch and Rivera 2017](#)), suggesting that apparent contagion often reflects common exposure rather than direct transmission. Whether the same pattern holds for labor strikes has not been established. The primary constraint has been data. Strike waves have been analyzed almost exclusively using aggregate country-level measures, which can establish correlations with economic conditions but cannot separate geographic transmission from national synchronization. Subnational panel data with sufficient geographic resolution to estimate neighbor effects while absorbing national shocks has not previously existed for labor strikes at global scale.

This paper provides the first direct test of geographic versus national strike diffusion using a new global dataset and a panel identification strategy that isolates geographic neighbor effects from national common shocks. The empirical analysis uses a validated subnational panel of first-level administrative units (ADM1) observed weekly from 2015 to 2025. The underlying source data come from the Global Database of Events, Language and Tone (GDELT), which provides event-level records with subnational geocoding across 181 countries through automated processing of global news text (Leetaru and Schrodtt 2013). Raw GDELT records for labor strikes contain substantial false positives and location errors, which I address through a two-stage LLM-based validation pipeline covering both event content and subnational geocoding. After extraction, filtering, and validation, the analysis sample contains 11,503 strike events geocoded to 568 ADM1 regions across 74 countries, yielding what is, to my knowledge, the first systematic global subnational panel of validated labor strike events.

I estimate two-way fixed effects models that pair unit fixed effects, absorbing time-invariant regional characteristics, with country-by-week fixed effects, which absorb all national-level common shocks simultaneously without requiring the researcher to specify which national factors matter. This design asks whether geographic neighbor strike exposure predicts strike occurrence in a focal region above and beyond what is explained by every national dynamic occurring in the same week. If geographic contagion is the primary mechanism, exposure to a striking neighbor should remain predictive even after national shocks are removed. If national contagion is the primary mechanism, the geographic association should disappear once national dynamics are controlled.

The results of the analysis are clear and point unambiguously toward the importance of national rather than regional geographic contagion. Regions with a striking neighbor are 3.5 times more likely to experience a strike in the raw data, a pattern visible in contemporary GDELT records that mirrors what historical accounts have described for more than a century. Yet this association fully disappears once country-by-week fixed effects are included. When geographic and national channels are estimated simultaneously, national contagion is large, robust, and explains orders of magnitude more within-unit variance. A one-standard-deviation increase in a country's strike share nearly doubles the focal region's strike probability. Geographic neighbor exposure, by contrast, is null net of national activity and turns slightly negative in the full decomposition, suggesting a mild displacement dynamic rather than geographic diffusion. Cross-border neighbor exposure is also null across all specifications. Strike contagion, unlike civil conflict spillovers, is bounded by national institutional frameworks rather than geographic proximity.

The analysis also tests whether institutional context conditions the spatial structure of strike contagion. The theoretical framework predicts that freedom of expression and freedom of association should have opposing effects on the geographic organization of strike waves. Where freedom of expression is stronger, strike signals can circulate more rapidly through national media, reducing the importance of geographic proximity. Where freedom of association is stronger, locally embedded union and associational networks should reinforce subnational clustering by transmitting mobilization through organizational ties that remain territorially concentrated. The results support these expectations. Freedom of expression weakens the geographic concentration of strike waves, while freedom of association strengthens it. Institutions condition the spatial distribution of strike waves without altering the strength of national contagion itself. The implication is that institutions shape not whether national strike waves occur, but how those waves are distributed across space.

These findings point to a broader research agenda. If strike contagion is primarily national, the next task is to explain the sources of national synchronization, including the economic shocks, political conflicts, organizational decisions, and bargaining cycles that generate country-wide waves of labor unrest. The analysis also directs attention to the mechanisms through which institutions shape the geography of those waves, particularly the roles of media systems, union organization, and other forms of associational infrastructure. It further underscores the importance of heterogeneity across strike types. General strikes, political strikes, and sector-specific industrial disputes likely follow different diffusion logics, and aggregate analyses may conceal important differences in how distinct forms of labor contention spread. More broadly, the paper argues that the geography of strikes cannot be understood apart from the national institutions and mobilizing processes that organize collective action across space.

The Spatial Structure of Strike Waves

The historical patterns documented in the comparative labor literature persist in contemporary cross-national data. Figure 1a plots weekly strike prevalence as the number of ADM1 regions recording more than one strike event per week in the GDELT database.¹ The smoothed trend reveals that, rather than occurring at a steady rate, strike activity concentrates into distinct episodes in which the number of regions simultaneously experiencing strikes rises well above the baseline.

¹Section 4 discusses the methods used to extract these data in detail.

Figure 1b shows the prevalence of strikes among regions with and without an actively striking neighbor in the prior two weeks. Regions with an actively striking neighbor experienced strikes at a rate of 4.9%, compared to 1.4% for unexposed regions, a 3.5-fold difference. The phenomenon that scholars have traced across industrial societies for more than a century remains visible, at global scale, in the present day.

What produces this clustering is less clear. The same observational patterns are consistent with two distinct causal accounts: one in which strikes spread through direct transmission between neighboring regions, and one in which workers across regions respond independently to shared national stimuli without directly influencing one another. The patterns in Figure 1 are identical under both mechanisms, and distinguishing between them requires both a theoretical argument about which is more likely to govern labor strikes and an empirical strategy capable of separating them.

I distinguish two mechanisms that could produce the patterns in Figure 1. *Geographic contagion* is a step-by-step transmission process in which strikes spread between spatially adjacent units through direct demonstration effects. Workers observe a neighboring region striking and update their beliefs about the feasibility and likely costs of collective action, lowering the threshold for action in their own region (Biggs 2002; Granovetter 1978). *National contagion* is coordinated mobilization driven by common national stimuli rather than sequential geographic diffusion. Union federations, political events, and national media simultaneously shift strike propensity across all regions, producing apparent clustering without any direct transmission between them.

McAdam, Tarrow, and Tilly (2008) provide a precise vocabulary for this distinction, contrasting *relational* mechanisms, in which transmission runs through direct ties between actors, with *non-relational* mechanisms, in which actors respond independently to shared stimuli. Geographic contagion is relational while national contagion is non-relational. As Oliver and Myers (2003) show, empirical models that cannot separate these two processes will systematically overestimate geographic diffusion effects. I argue that for labor strikes, national contagion is the predominant mechanism. I further hypothesize that the relative weight of the two mechanisms varies with institutional context, with the structure of information environments and associational freedoms conditioning the spatial structure through which national mobilization propagates.

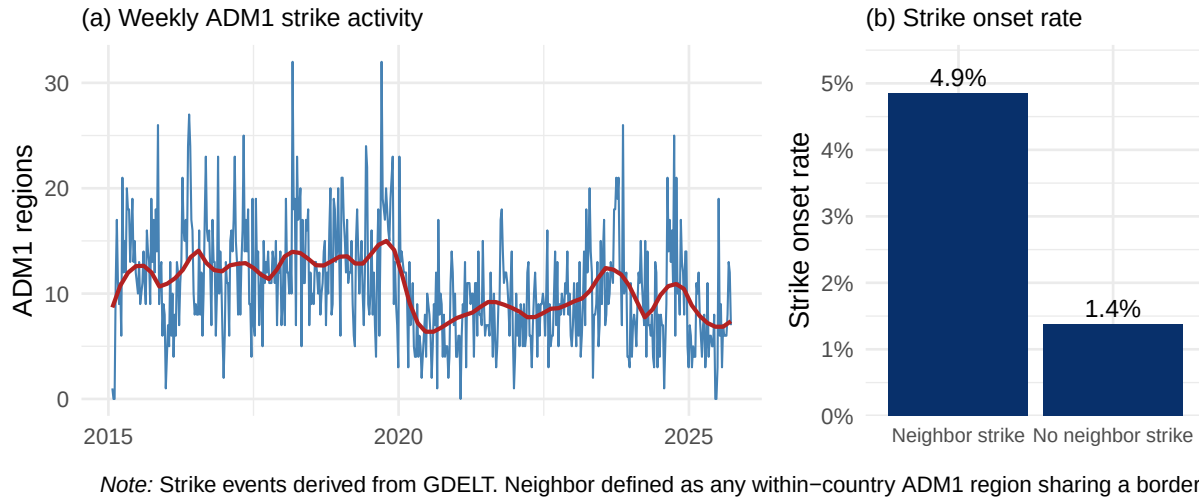


Figure 1: Strike activity and neighbor exposure.

The National Foundations of Strike Waves

The comparative-historical literature on strike waves consistently finds that elevated strike activity sweeps multiple regions and industries simultaneously rather than radiating outward from a geographic source. Shorter and Tilly's (1974) analysis of French strike waves, Cronin's (1987) documentation of the great British waves of 1871 to 1920, and Franzosi's (1989, 1995) analyses of Italian strikes all conclude that major waves are nationally synchronized, driven by macro-level political and economic cycles that create the same mobilization moment across regions at once. Strike waves are synchronized in part because the relevant information, about employer resistance and political opportunity, travels through national media and union networks rather than through geographic proximity (Biggs 2002). Broad shifts in political opportunity generate mobilization across sectors and regions simultaneously, producing spatial clustering as a byproduct of common national exposure rather than geographic spread (S. G. Tarrow 2011).

Economic theories of strikes reinforce the primacy of national dynamics from a different perspective. In the canonical bargaining model of strikes, strike probability is a function of the gap between worker expectations and firm capacity to pay, a gap driven by macroeconomic conditions that are national in scope (Ashenfelter and Johnson 1969; Kennan and Wilson 1993; Card 1990). When unexpected shifts in profitability, inflation, or real wages affect all firms in a country simultaneously, strike timing will be synchronized across regions not because workers are imitating each other but because all face the same economic environment. This synchronization has been

documented empirically at the macro level (Rees 1952) and extended cross-nationally, with the late-1960s and early-1970s strike waves across France, Italy, Britain, and the United States attributed to shared macroeconomic and political conditions rather than sequential contagion (Hibbs 1976, 1978).

The Limits of Geographic Contagion

An established literature on the spatial diffusion of collective action finds that geographic proximity predicts contagion, but largely for violent and civil conflict rather than for labor organizing. Robust neighborhood effects that survive structural controls have been documented in studies of civil war onset (Buhaug and Gleditsch 2008), with regional conflict environments shaping civil war onset independently of domestic conditions (Kristian Skrede Gleditsch 2007). The mechanisms in these cases are local in character, including rebel group expansion across borders, refugee flows that import instability, ethnic networks spanning administrative boundaries, and competition over natural resources that straddle frontiers.

None of these mechanisms translate straightforwardly to labor strikes. The organizational infrastructure that translates mobilization signals into strike action, including labor unions, collective bargaining agreements, and labor law, operates overwhelmingly at the national level (Shorter 1974; Franzosi 1995). A union federation's decision to call a general strike, or a government's announcement of austerity measures, creates the same organizational moment for workers in every region simultaneously. Geographic adjacency between administrative units adds little to this national signal.

Even in the nonviolent protest literature, geographic neighbor effects are weaker and more conditional than in conflict studies. The geographic diffusion of nonviolent campaigns has been shown to depend on shared political opportunity structures, with spatial proximity contributing little once those structures are controlled (Kristian S. Gleditsch and Rivera 2017). Genuine geographic diffusion requires active organizational brokerage, with new ties deliberately constructed to link previously isolated actors, and without such linkages what appears to be geographic spread is better understood as simultaneous response to a common environment (S. Tarrow and McAdam 2005).

Institutions and the Geography of Strike Waves

Even if national contagion predominates on average, the spatial distribution of nationally-driven mobilization may vary with the institutional context in which workers and employers operate. I consider three dimensions of institutional context that may condition how strike signals travel: freedom of expression, freedom of association, and state repression.

Freedom of expression shapes the information infrastructure through which national strike signals reach workers. Where press freedom is strong and media access is broad, information about strikes travels through national broadcast channels, reaching workers in all regions simultaneously regardless of their geographic position. Collective violence has been shown to diffuse along media distribution networks rather than through geographic proximity ([Myers 2000](#)), and the same logic applies to strike news. Where national media delivers the mobilization signal everywhere at once, no region needs a striking neighbor to be activated, and the geographic structure of strike waves should be correspondingly attenuated.

Freedom of association shapes the organizational infrastructure through which workers translate national mobilization signals into local action. Where associational rights are strong, dense union networks and civil society organizations sustain direct linkages between workers, but those networks are often locally embedded, organized at the plant, district, or regional level rather than exclusively through national federations. Where such locally structured networks are the primary conduit for strike information and coordination ([Biggs 2002](#)), geographic proximity becomes more rather than less important for how the national signal propagates. Higher freedom of association should therefore amplify the geographic structure of strike waves, producing more spatially clustered national episodes.

State repression raises the costs of collective action and should suppress the translation of strike signals into local mobilization. Where states have the capacity and will to suppress collective action, workers may observe strike activity nearby or perceive a national mobilization wave and nonetheless not act, because the expected costs of participation outweigh the informational update from the strike signal. This logic has been demonstrated for conflict contagion ([Braithwaite 2010](#)), and the parallel argument for strikes suggests that higher repression should dampen both geographic and national transmission.

Theoretical Expectations

The arguments developed above yield five expectations, two concerning the main contagion effects and three concerning institutional conditioning. National contagion should be positive and substantial, with the probability that any given region experiences strike activity rising as a greater share of a country's regions are already mobilized, reflecting the common national mobilization environment. Geographic contagion, by contrast, should be negligible once national dynamics are controlled, because spatial proximity between administrative units adds little explanatory power beyond the national signal that is the primary conduit for strike transmission.

The institutional conditioning arguments generate three further expectations. Higher freedom of expression should attenuate the geographic neighbor effect, as broadcast media substitutes for geographic proximity as the primary conduit for strike signals. Higher freedom of association should amplify it, as locally embedded organizational networks reinforce the role of proximity in transmitting mobilization. The opposite signs of these two predictions reflect a single coherent theoretical claim: the two institutions route the same national mobilization signal through different infrastructures, one that flattens geographic structure and one that reinforces it. Higher repression should dampen the translation of strike signals into local mobilization through both channels.

Strike Data and Validation

Testing the theoretical expectations requires systematic data on strike activity at subnational resolution across a large and diverse set of countries over an extended period. No existing source has met these requirements. The International Labour Organization publishes annual strike statistics compiled from national labor ministry reports, but these aggregate to the country-year level and contain no event-level or subnational geographic information. The High Profile Strikes Dataset ([Robertson and Teitelbaum 2011](#)) provides cross-national event records of major industrial actions and represents the most comprehensive global strike compilation assembled to date, but it is restricted to high-salience cases and is no longer in production. Broader conflict and protest databases offer event-level geocoded records but do not systematically cover labor strikes as a distinct event type. ACLED explicitly excludes strikes from its protest category unless they are accompanied by a public demonstration ([Raleigh et al. 2010](#)), rendering coordinated work stoppages invisible in the data.

This analysis draws on the Global Database of Events, Language and Tone (GDELT), which

offers event-level records with subnational geocoding across a large number of countries and includes a dedicated identifier for strike events (Leetaru and Schrodtt 2013). GDELT identifies events through automated processing of news text, a design that maximizes global coverage but generates substantial false positives and introduces location errors at the subnational level. The remainder of this section describes the steps taken to extract and filter the data, the validation pipeline used to address these problems, the construction of the analysis panel, and the key variables used in the analysis.

Strike Events and False Positives

GDELT codes events using the Conflict and Mediation Event Observations (CAMEO) scheme, which assigns each reported event to one of several hundred action categories (Leetaru and Schrodtt 2013; Schrodtt 2012). The relevant category here is CAMEO code 143, labeled “strikes and boycotts,” which covers labor strikes, work stoppages, and political boycotts. I queried the GDELT database via Google BigQuery for all code-143 events from January 2015 to September 2025, retaining only records with valid geographic coordinates, which yielded 492,500 records. Three further filters were then applied to improve data quality before any content-based validation. First, I kept only events assigned to a specific subnational administrative region rather than to a country as a whole, reducing the dataset to 328,982 records. Then I filtered for events with at least one high-confidence news mention, where confidence reflects how closely the source article matches the coded event according to GDELT’s internal scoring, leaving 134,141 records. Then I filtered for events where the associated news articles were tagged with the keyword “STRIKE” in GDELT’s article-level topic metadata, reducing the dataset to 30,899 records.

Table 1: Data extraction and filtering pipeline.

Step	Description	Records
BigQuery download	CAMEO code 143, valid coordinates	492,500
Geographic filter	Valid subnational region code assigned	328,982
High-confidence filter	At least one high-confidence news mention	134,141
Keyword theme filter	Article tagged "STRIKE" in topic metadata	30,899
LLM content classifier	Article describes a labor strike	11,503

Note: Each row shows the number of records retained after applying the listed filter.

Even after applying these filters, the dataset retains a substantial number of false positives. GDELT’s design prioritizes capturing as many events as possible rather than ensuring each record reflects a real event, and raw records include misclassified articles, retrospective coverage of past events treated as current, and articles that mention strikes only in passing. The CAMEO coding scheme also groups labor strikes together with political boycotts under a single event type. Applying CAMEO code 143 directly as a measure of labor strikes would therefore mix genuine industrial actions with other forms of collective action for which the contagion mechanisms described in Section 2 need not apply.

LLM Validation Pipeline

Due to the false positive issue identified above, previous studies using GDELT have heavily filtered the data. Wang et al. (2016) analyzed the URLs underlying raw GDELT protest event records and found that only 21% of valid URLs actually described a protest event. Hoffmann et al. (2022) applied a machine learning classifier to GDELT protest events in six European countries, retaining 3,564 validated records from 379,747 raw events, with precision of 0.74 and recall of 0.83. For GDELT to serve as a credible source for subnational strike analysis, a comparable filtering step is required. I apply a two-stage validation pipeline that addresses these problems in sequence. The first stage uses a large language model to assess whether each remaining record actually describes

a labor strike. The second checks whether the location GDELT has assigned to each validated event is correct.

Stage 1: URL Classifier

The first stage uses GPT-4o mini to assess whether each of the 30,899 remaining records actually describes a labor strike, following the approach described by Halterman and Keith (2025). The model is prompted using few-shot prompting whereby the prompt includes a set of labeled examples that demonstrate the desired classification before the model is asked to classify a new case. Each prompt pairs the source URL and headline with a system instruction defining what counts as a labor strike and a series of worked examples covering relevant, irrelevant, and ambiguous cases across multiple languages. The model is then asked whether the underlying article covers a labor strike rather than a related but distinct event such as a political boycott. Full article text retrieval proved unnecessary, since news coverage of industrial disputes almost invariably names the location and nature of the dispute in the headline or web address. Uncertain cases from the first pass are resubmitted in a second pass with the article headline added when it can be retrieved. Unlike Hoffmann et al.'s (2022) classifier, which involved full article retrieval and was limited to six countries, this approach scales to the full global dataset without additional infrastructure. The classifier retained 11,503 events from the 30,899 records submitted. The full system prompt and few-shot examples for both passes are reproduced in the appendix.

Stage 2: Geocode Validator

The second stage checks whether GDELT's subnational location assignment is correct for each validated event. GDELT determines location by scanning article text for place names, a process that introduces systematic errors at the subnational level (Leetaru and Schrodt 2013). The most consequential, documented by Hammond and Weidmann (2014), is dateline bias: when a wire service or foreign correspondent covers a strike in a secondary city, the dispatch location rather than the strike location can be recorded as the event's geographic origin.

I address these errors with a second model pass using GPT-5 mini. As in Stage 1, the model is prompted with few-shot examples illustrating each type of location error before being asked to assess each case. It receives the source URL and headline and is asked whether GDELT's recorded location is consistent with the geographic content of the article. The full system prompt and examples are reproduced in the appendix. Among the 11,503 validated events, the model returned

a definitive verdict for 5,236 events. The remaining 6,267 received an uncertain verdict, predominantly because the source URL returned a broken link or paywalled content that provided too little information to judge. Among events with a definitive verdict, 34.5% were miscoded (1,806 corrected; 3,430 confirmed). Where the model identified a more accurate location, that corrected assignment replaces GDELT's original. Country-level error rates range from under 10% to over 90%, revealing the systematic nature of the problem.² Each validated event is assigned a quality indicator reflecting whether its location was confirmed, corrected, or left unvalidated, and this indicator is used in robustness checks that restrict the sample to location-verified observations.

Constructing the ADM1-Week Panel

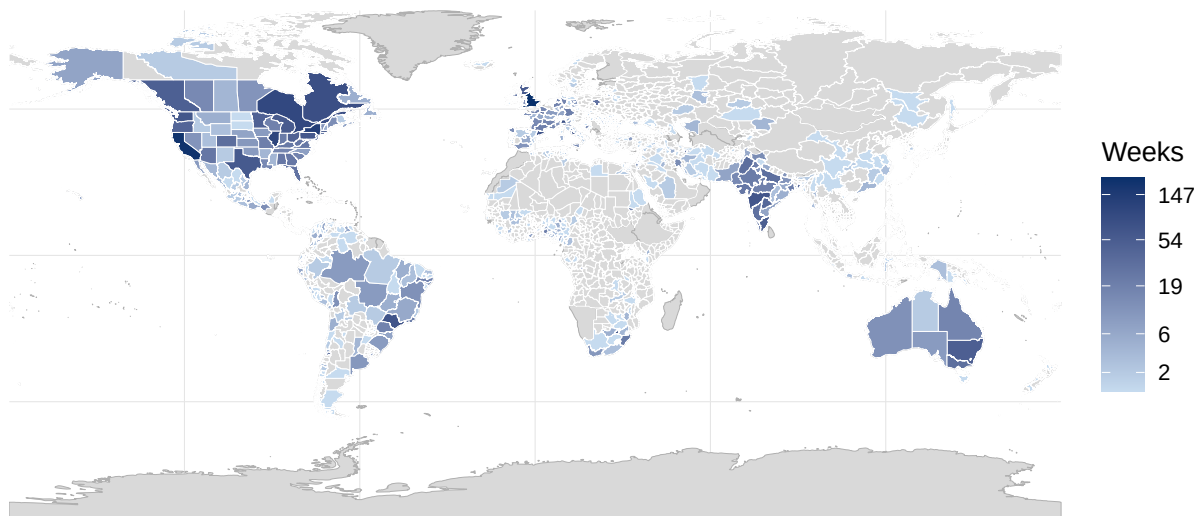
I aggregate validated strike events to an administrative region-week panel spanning January 2015 to September 2025. The outcome variable is a binary strike occurrence indicator equal to one if at least one validated event was recorded in the region during week t . The baseline occurrence rate is 1.4% across all region-weeks (0.87% in the subsample restricted to location-verified events).

The panel is restricted to administrative regions in which at least one validated strike event was recorded during the sample period. In media-derived event data, a zero for a region that never appears in the validated sample is difficult to interpret, because it may reflect true non-occurrence or systematic non-coverage. Treating all such regions as true zeros would introduce substantial measurement error correlated with media visibility, which varies widely across subnational units and is especially severe in rural areas, low-coverage media markets, and settings where reporting is sparse or difficult to geocode. The analysis therefore focuses on regions for which the data provide evidence of meaningful observability in the news stream relevant to strike reporting. The resulting estimates should be interpreted as applying to subnational units with sufficient media presence to enter the validated GDELT strike sample, rather than to all administrative regions in the world.

After restricting to regions with at least one within-country neighbor also present in the validated data, the analysis panel includes 568 regions across 74 countries, observed over 556 weeks, yielding 315,808 region-week observations. Figure 2 maps cumulative strike prevalence across regions, showing concentration in South Asia, sub-Saharan Africa, and parts of Europe, reflecting

²Error rates are highest for China (95%), Russia (77%), Ukraine (74%), Mexico (70%), and Israel (64%), almost certainly reflecting outlet-based location assignment: English-language coverage of strikes in these countries is disproportionately written by foreign correspondents, and GDELT records the correspondent's location rather than the strike site. Canada (49%) and the United Kingdom (39%) also show elevated rates, likely reflecting subnational misassignment within English-speaking countries.

both genuine geographic variation and GDELT’s media coverage patterns.³



Note: Strike events derived from GDELT. Administrative boundaries from GADM v4.1.

Figure 2: Cumulative strike prevalence across ADM1 regions, 2015–2025.

Defining Geographic Neighbors

I define geographic neighbors as administrative regions sharing a border within the same country, identified using GADM version 4.1 administrative boundary polygons. The main analysis uses within-country neighbors only. Cross-border adjacency is examined as an exploratory extension. Table A1 (appendix) summarizes the distribution of neighbor counts.

Outcome, Treatment, and Moderators

Geographic Neighbor Exposure

The primary treatment variable, `neighbor_strike_within_t1_t2`, is a binary indicator equal to one if at least one within-country neighboring region recorded a strike occurrence in week $t - 1$ or $t - 2$. Exposure in either of the two prior weeks counts as exposure. The two-week window is designed to capture the period over which a demonstration effect from a neighboring region’s strike might plausibly influence workers’ decisions in an adjacent region, while keeping

³GDELT’s event detection is based on automated processing of global news text and coverage is stronger for sources using romanized scripts. English-language and Western newswire sources are substantially better represented than sources in non-romanized languages such as Chinese, Arabic, Japanese, or Korean. Countries with active English-language domestic media or significant foreign bureau presence — particularly in South Asia and sub-Saharan Africa — are therefore overrepresented relative to their actual strike activity, while East Asia and much of the Middle East are underrepresented. This geographic selectivity is a known limitation of media-derived event data (Leetaru and Schrodtt 2013) and is part of what Figure 2 reflects.

the treatment variable predetermined with respect to the current week's outcome. Only 7.9% of region-weeks are exposed, reflecting the sparsity of the validated strike sample.

National Contagion

For the decomposition specification, `country_share_t1_t2` is the average share of regions in the same country with a strike occurrence in $t - 1$ and $t - 2$, normalized by $(N_c - 1)$ to remove any mechanical relationship between the variable's scale and country size.

Institutional Moderators

Three country-year indicators from the Varieties of Democracy (V-Dem) project are used in the moderation models. These are freedom of expression (`v2x_freexp_altnf`), freedom of association (`v2x_frassoc_thick`), and physical violence by state agents (`v2x_clphy`). All three are standardized before entering interaction models so that the main treatment coefficient is interpretable at the average value of each moderator in the sample.

Empirical Strategy

The analysis tests the theoretical expectations through three specifications. The first asks whether geographic neighbor exposure predicts strike occurrence once national co-movement is absorbed, using a two-way fixed effects design that pairs ADM1 unit fixed effects with country-by-week fixed effects. The country-by-week fixed effects are the key identifying restriction, absorbing all within-country variation in a given week without requiring the researcher to specify which national factors matter. A second specification estimates geographic and national contagion simultaneously by replacing the country-by-week fixed effect with an explicit measure of country-level strike activity, allowing both effects to be compared directly in the same model. A third specification then interacts the treatment variables with V-Dem indicators of freedom of expression, freedom of association, and state repression to test whether institutional context conditions the spatial structure of diffusion.

Two-Way Fixed Effects Model

To test whether geographic neighbor exposure predicts strike occurrence once common national dynamics are removed, I estimate:

$$Y_{it} = \beta \cdot \text{NeighborStrike}_{i,t-1:t-2} + \alpha_i + \gamma_{c \times t} + \varepsilon_{it}$$

where Y_{it} is the binary strike occurrence indicator for ADM1 unit i in week t , $\text{NeighborStrike}_{i,t-1:t-2}$ is a binary indicator equal to one if at least one within-country geographic neighbor recorded a strike occurrence in week $t - 1$ or $t - 2$, α_i is an ADM1 fixed effect, $\gamma_{c \times t}$ is a country-by-week fixed effect for country c in week t , and ε_{it} is the error term. I use two lags to capture a plausible demonstration-effect window while keeping the treatment predetermined with respect to the current period's outcome.

I estimate this as a linear probability model (LPM) via `feols` in R (Bergé 2018). LPM is preferred over logit in this setting because maximum likelihood estimators with many fixed effects suffer from the incidental parameters problem, which does not affect OLS (Lancaster 2000; Wooldridge 2010). Coefficients are directly interpretable as marginal effects on the probability of strike occurrence. Given the low baseline occurrence rate of 1.4%, I report a bias-corrected logit robustness check using the `alpaca` package (Fernández-Val and Weidner 2016) (see Table A2). The substantive conclusions are unchanged.

Standard errors are clustered at the ADM1 level to account for serial correlation within units over time. Ignoring within-unit serial correlation in panel settings leads to substantially understated standard errors (Bertrand, Duflo, and Mullainathan 2004), and clustering at the ADM1 level is appropriate because geographic neighbor exposure varies at that level (Colin Cameron and Miller 2015).

The country-by-week fixed effect represents a more demanding identification strategy than is typical in the conflict contagion literature, which generally uses logit with parametric controls for national conditions such as democracy, GDP per capita, and population (Buhaug and Gleditsch 2008; Kristian S. Gleditsch and Rivera 2017). Rather than specifying which national factors to control for, the saturated fixed effect absorbs all of them simultaneously. This approach has been standard in the international trade literature, where country-by-time fixed effects absorb unobservable common shocks without requiring the researcher to model their form (Baier and Bergstrand 2007; Head and Mayer 2014). The lagged-treatment design also has methodological precedent in the study of historical collective action: Aidt, Leon, and Satchell (2022) and Aidt and Leon (2022) employ a similar subnational spatial panel approach to study the geographic diffusion of the English Swing Riots of 1830–31. The present study extends this design to contemporary global data

and, unlike the Swing Riots case, finds no geographic contagion once national dynamics are controlled. An alternative framework would be an explicit spatial lag model, as in Arezki et al. (2024), who study protest contagion across 200 countries using a spatial autoregressive specification; however, the lagged-treatment TWFE approach is preferred here because the predetermined neighbor indicator avoids the simultaneity problem inherent in contemporaneous spatial lags.

National vs. Geographic Decomposition

The central argument of the paper is that strike contagion operates at the national rather than geographic level, and that the raw geographic association in the data reflects common national dynamics rather than direct transmission between neighboring regions. Testing this argument requires estimating national and geographic contagion simultaneously. I therefore replace the country-by-week fixed effect with an explicit measure of country-level strike activity, constructing a specification that places both channels in the same model:

$$Y_{it} = \beta_1 \cdot \text{NeighborStrike}_{i,t-1:t-2} + \beta_2 \cdot \text{CountryShare}_{i,t-1:t-2} + \alpha_i + \gamma_t + \varepsilon_{it}$$

where $\text{CountryShare}_{i,t-1:t-2}$ is the proportion of other ADM1 regions in the same country with a strike occurrence in weeks $t - 1$ or $t - 2$ normalized by $(N_c - 1)$ to remove any mechanical relationship with country size, α_i is an ADM1 fixed effect, γ_t is a week fixed effect common to all countries, and ε_{it} is the error term. This specification estimates national contagion (β_2) and the geographic neighbor effect net of national activity (β_1) simultaneously. > This specification replaces the country-by-week fixed effect with an explicit measure of national strike activity, allowing the national and geographic channels to be estimated separately. If the raw geographic association reflects national rather than geographic dynamics, as the theoretical framework predicts, β_2 should be large and positive while β_1 collapses toward zero.

Testing Institutional Conditioning

The third set of expectations concerns whether institutional context conditions the spatial structure of diffusion. To test these, I interact both the geographic neighbor treatment and the country-share measure with V-Dem indicators:

$$\begin{aligned}
Y_{it} = & \beta_1 \cdot \text{NeighborStrike}_{i,t-1:t-2} + \beta_2 \cdot \text{CountryShare}_{i,t-1:t-2} + \beta_3 \cdot \text{Institution}_{ct} \\
& + \beta_4 \cdot (\text{NeighborStrike}_{i,t-1:t-2} \times \text{Institution}_{ct}) \\
& + \beta_5 \cdot (\text{CountryShare}_{i,t-1:t-2} \times \text{Institution}_{ct}) + \alpha_i + \gamma_t + \varepsilon_{it}
\end{aligned}$$

where Institution_{ct} is the relevant country-level institutional measure for country c at time t , α_i is an ADM1 fixed effect, γ_t is a week fixed effect, and ε_{it} is the error term. These specifications draw on cross-national variation in institutional context to test whether the structure of diffusion varies with the political environment in which workers operate. I use ADM1 and week fixed effects, include CountryShare as an explicit measure of national waves, and allow institutional context to condition both the geographic and national channels. The moderation effects are identified primarily from between-country differences in institutional context. The estimates are therefore best read as descriptive of cross-national heterogeneity rather than within-country causal moderation. All V-Dem indicators are standardized so that the main coefficients on NeighborStrike and CountryShare are interpreted at the sample mean of each moderator.

Main Results

The analysis proceeds in three stages. First, I estimate the baseline effect of geographic neighbor strike exposure using a two-way fixed effects strategy that absorbs common national shocks. Second, I decompose the apparent neighborhood effect into national and geographic channels. Third, I examine whether institutional context conditions the geographic structure of strike contagion.

Neighbor Effects Disappear with National Controls

Table 1 presents estimates from the baseline two-way fixed effects model in Equation (1). Column (1) uses the pooled two-lag neighbor measure, while columns (2) and (3) replace it with separate indicators for exposure at $t-1$ and $t-2$. Geographic neighbor strike exposure does not significantly predict strike occurrence once common within-country shocks are absorbed, a striking result given the raw association (see Figure 1). Country-by-week fixed effects fully absorb this association, indicating the raw correlation reflects common national dynamics rather than geographic diffusion.

Table 2: Baseline specification: neighbor strike exposure and strike occurrence

Model:	strike_onset		
	(i)	(ii)	(iii)
<i>Variables</i>			
Neighbor strike (t−1, t−2)	-0.0031 (0.0021)		
Neighbor strike (t−1 only)		-0.0058** (0.0027)	
Neighbor strike (t−2 only)			-0.0014 (0.0023)
<i>Fixed-effects</i>			
ADM1	Yes	Yes	Yes
Country × Week	Yes	Yes	Yes
<i>Fit statistics</i>			
Observations	315,808	315,808	315,240
R ²	0.19665	0.19668	0.19670
<i>Clustered (ADM1) standard-errors in parentheses</i>			
<i>Signif. Codes: ***: 0.01, **: 0.05, *: 0.1</i>			

Examining the two lags separately sharpens the picture. The t−1 lag is negative and marginally significant ($\beta = -0.006$, $p = 0.036$), while the t−2 lag is null. Rather than pointing to a modest diffusion effect at one lag, the negative t−1 coefficient suggests a displacement or substitution dynamic in which the focal region becomes *less* likely to follow immediately once a neighboring region strikes, a pattern discussed further below.

National Contagion Dominates Neighbor Effects

Table 2 presents estimates from the decomposition model in Equation (2), which replaces the country-by-week fixed effect with the explicit country strike share measure. Column (1) includes the national channel alone, column (2) includes the geographic neighbor channel alone, and column (3) estimates both simultaneously. Column (1) establishes that national-level contagion is large and highly significant. A one percentage-point increase in the country strike share

raises the focal region's strike probability by 0.66 percentage points; at a one-standard-deviation increase, this represents a near-doubling of occurrence risk. National contagion alone explains 6.3% of within-unit variance, a substantial share given the sparseness of the outcome.

Table 3: National vs. neighborhood decomposition

Model:	strike_onset		
	(i)	(ii)	(iii)
<i>Variables</i>			
Country share (t-1, t-2)	0.6620*** (0.0501)		0.6653*** (0.0507)
Neighbor strike (t-1, t-2)		0.0149*** (0.0018)	-0.0043** (0.0020)
<i>Fixed-effects</i>			
ADM1	Yes	Yes	Yes
Week	Yes	Yes	Yes
<i>Fit statistics</i>			
Observations	315,808	315,808	315,808
R ²	0.13925	0.08181	0.13931
<i>Clustered (ADM1) standard-errors in parentheses</i>			
<i>Signif. Codes: ***: 0.01, **: 0.05, *: 0.1</i>			

Column (2) isolates the geographic neighbor effect. In the absence of controls for national activity, having a striking neighbor is positive and significant (+1.5 percentage points, $p < 0.001$), consistent with prior literature on geographic diffusion.

However, column (3) is the key specification. When both measures are included simultaneously, the national share coefficient is unchanged (+0.665, $p < 0.001$) while the geographic neighbor effect reverses sign ($\beta = -0.004$, $p = 0.035$). Once national activity is controlled, having a geographically adjacent neighbor strike slightly reduces the focal region's strike probability.

The contrast in explanatory power is stark. National contagion explains 63 times more within-unit variance than geographic neighbor exposure (within $R^2 = 0.063$ versus 0.001). The 1.5 percentage-point neighborhood effect in column (2) is entirely attributable to national wave dynamics. Workers in the same country tend to strike together not because geographic proximity

transmits the mobilization signal, but because a common national dynamic drives coordinated action.⁴

Institutions Shape the Geography of National Waves

Table 3 presents estimates from the institutional conditioning model in Equation (3). The table interacts both the neighbor strike measure and the country share measure with three V-Dem indicators, entered individually in columns (1) through (3) and jointly in column (4), while retaining ADM1 and week fixed effects.

⁴The negative coefficient on the geographic neighbor indicator should be interpreted with caution. It is identified from the narrow set of cases where a region's neighbors diverge from national trends, a relatively rare occurrence by construction, and is therefore estimated from thinner variation than the country share coefficient.

Table 4: Institutional moderation of geographic and national contagion

Model:	strike_onset			
	(i)	(ii)	(iii)	(iv)
<i>Variables</i>				
Neighbor strike (t-1, t-2)	-0.0047** (0.0021)	-0.0060*** (0.0020)	-0.0046** (0.0020)	-0.0067*** (0.0020)
Freedom of expression (z)	0.0001 (0.0015)			0.0004 (0.0029)
Country share (t-1, t-2)	0.6544*** (0.0496)	0.6507*** (0.0489)	0.6588*** (0.0496)	0.6507*** (0.0493)
Neighbor strike × Expression	0.0018 (0.0022)			-0.0121** (0.0054)
Country share × Expression	0.0551 (0.0469)			0.0561 (0.1535)
Freedom of association (z)		-0.0004 (0.0019)		-0.0016 (0.0028)
Neighbor strike × Association		0.0045** (0.0023)		0.0157*** (0.0058)
Country share × Association		0.0618 (0.0477)		0.0542 (0.1351)
State repression (z)			0.0005 (0.0013)	0.0012 (0.0020)
Neighbor strike × Repression			0.0024 (0.0021)	0.0010 (0.0024)
Country share × Repression			0.0295 (0.0451)	-0.0507 (0.0774)
<i>Fixed-effects</i>				
ADM1	Yes	Yes	Yes	Yes
Week	Yes	Yes	Yes	Yes
<i>Fit statistics</i>				
Observations	294,224	294,224	294,224	294,224
R ²	0.14130	0.14136	0.14109	0.14151

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Clustered (ADM1) standard-errors in parentheses

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Institutional context does not condition how strongly national strike waves propagate. The country share interactions are null across all three moderators and in the joint model, and any conditioning operates exclusively on the geographic channel. When modeled jointly, freedom of expression attenuates the neighbor strike effect ($\beta = -0.009$, $p = 0.073$) while freedom of association amplifies it ($\beta = +0.010$, $p = 0.062$), with effects that are nearly equal in magnitude and opposite in sign. Repression is null throughout. When modeled separately, these effects are attenuated because freedom of expression and freedom of association are positively correlated and their opposing effects partially cancel when either is included in isolation.

In countries with strong freedom of expression, information about strikes spreads rapidly through national media, enabling workers in distant regions to coordinate simultaneously rather than through geographic proximity, producing nationally homogeneous strike waves with little geographic gradient. In countries with strong freedom of association, dense union networks are locally embedded at the plant, district, or regional level, creating organizational infrastructure that reinforces the role of geographic proximity in transmitting mobilization signals, producing geographically structured national episodes rather than uniform waves. In short, freedom of expression produces nationally homogeneous strike waves while freedom of association produces geographically structured ones.

These results must be qualified by the decomposition finding. Since the geographic neighbor effect approaches zero and turns negative once country-level contagion is controlled, the moderation results are best understood as capturing the spatial structure of nationally-driven mobilization rather than true neighborhood diffusion. Institutions shape how national strike waves distribute across space but do not appear to generate or amplify geographic transmission as an independent mechanism.

Robustness Checks

The null geographic contagion result is robust across alternative specifications. Two-way clustering and logistic regression leave the main findings unchanged (Table A2). Results also replicate in the validated-location subsample, which restricts to strike events with confirmed ADM1 geocodes (Tables A3 and A4), with the null geographic neighbor effect and large national contagion both intact. Cross-border neighbor exposure is null across all specifications (Table A5), consistent with the view that strike contagion is bounded by national institutional frameworks.

Implications and Extensions

The results establish three robust findings: geographic neighbor exposure has no independent effect on strike occurrence once national dynamics are controlled; national contagion is large and explains orders of magnitude more within-unit variance than geographic proximity; and civil liberties shape the spatial structure of nationally-driven mobilization without conditioning its overall strength. Together these findings suggest that future research should focus on understanding what produces national synchronization and how institutions determine whether that synchronization takes a geographically uniform or spatially clustered form. This discussion highlights three questions related to this task. What drives national strike waves and what would it take to study them? Why do institutional conditioning effects appear modest and difficult to pin down? And does the contagion process look different for different types of strikes?

Why Institutional Conditioning Is Hard to Pin Down

While the institutional moderation results are theoretically coherent, the estimates are limited by two key factors. The first is a fundamental measurement issue. GDELT's media-derived event detection is systematically thinner in high-repression environments. In countries where the state actively restricts press access, suppresses protest reporting, or controls media outlets, the observable strike record is likely a highly selected sample of actual labor activity, capturing only the most egregious, internationally newsworthy episodes rather than the full distribution of strike behavior that the theory targets. Repression both suppresses the mobilization the analysis is trying to detect and degrades the measurement instrument simultaneously. As a result, a null result in this context cannot be interpreted as a genuine null.

A second problem affects all three V-Dem moderators. Freedom of expression, freedom of association, and state repression are measured annually at the country level and move very slowly over time. As a result, identifying variation in the interaction terms is almost entirely cross-sectional and indicates whether countries with higher civil liberties show different geographic strike structures as opposed to whether a country's strike geography changes as its institutions evolve. ADM1 and week fixed effects cannot absorb the many ways that high-civil-liberty countries differ from low-civil-liberty countries, including levels of development, state capacity, democratic consolidation, and union density, so the moderation estimates should be interpreted as exploratory cross-national comparisons rather than causal estimates of institutional effects.

Resolving these problems requires measures with substantially more temporal variation. One promising direction draws on the same event-data infrastructure used to construct the strike panel itself. GDELT's Global Knowledge Graph codes thematic content of news articles including media censorship events, and actor-type codes identify labor organizations, civil society groups, and government coercion events. From these it is possible, in principle, to construct country-month or even country-week proxies for information environment (volume and source diversity of media coverage, frequency of censorship-related events) and associational density (frequency of labor organization and civil society mentions). ACLED provides event-level data on state violence against protesters at weekly resolution across more than 150 countries, which could serve as a dynamic repression measure that varies within country over time. The advantage of these alternatives is precisely that they would generate within-country temporal variation, allowing the moderation analysis to be identified from changes in institutional context within a country rather than from cross-country comparisons, and making it possible to absorb country-level unobservables in a way that the current approach cannot.

Freedom of expression and freedom of association are also positively correlated across countries, which limits the ability to identify their opposing effects when either enters the model in isolation. This is why the joint model is the appropriate specification, but even there the estimates are noisier than they would be if the two institutions were orthogonal. Direct measures of the underlying mechanisms, including media reach across subnational regions, local union density at the ADM1 level, or other measures of associational infrastructure, could provide sharper tests of the theoretical claims than annual national indices that bundle many distinct institutional features into a single score.

Explaining National Strike Waves

The two-way fixed effects design is conservative by construction: country-by-week fixed effects absorb all national common shocks, including the wave dynamics that the large national contagion effect implies are present and substantial. This conservatism is the right choice for the geographic contagion test, but it purchases identification at the cost of any information about what generates national waves in the first place. The current paper establishes that when national waves occur they produce large and spatially broad increases in strike probability, but what triggers them remains unexamined.

A natural extension of this research is a country-week or country-quarter model of aggre-

gate strike activity, asking what economic and political conditions predict entry into high-strike periods. The macroeconomic drivers of strike waves are well-theorized. Wage-price cycles, unexpected shifts in firm profitability, unemployment trajectories, and real wage growth have all been implicated in the historical wave literature, and these can be approximated from IMF and World Bank sources at quarterly frequency. The political opportunity literature points to government partisanship, election cycles, and austerity episodes as triggers for politically timed waves; here V-Dem and ParlGov provide the relevant variables. Union density and bargaining centralization from the OECD and ILO would capture the organizational infrastructure that converts economic grievances into coordinated action for at least a subset of countries. Contract cycle dynamics, whereby periods when many workers' agreements come up for renewal simultaneously create windows of elevated strike risk, could also be examined using ILO labor statistics where available, and GDELT-derived time series of labor negotiation events and wage-related news coverage could serve as leading indicators in countries where official statistics are sparse.

Strike Types and the Geography of Mobilization

A third question concerns whether the contagion process looks the same across different types of strikes. The current analysis treats all CAMEO 143 events as equivalent, aggregating general strikes, industrial strikes, and political protest strikes into a single onset indicator. This may obscure substantial heterogeneity in how different kinds of labor action spread, and may explain at least part of why geographic and national channels behave as they do in the aggregate.

The contrast between industry-specific and general strikes is theoretically sharp. Workers in particular industries are often geographically concentrated. Coal mines cluster in mining regions, garment factories in export processing zones, port workers in coastal cities, while public school teachers are distributed across every administrative region simultaneously. A wage dispute in a textile corridor may produce correlated strike activity in neighboring textile regions through genuine industry-network channels, including shared union federations, the same employer association, and synchronized contract cycles, while having no effect on non-textile regions in the same country. This pattern would look like geographic contagion in the raw data and would survive some national controls, even though the mechanism is sectoral rather than spatial. The current design cannot rule out that spatially clustered sectors account for the modest geographic residuals identified here. Disaggregating by industry would make this testable directly.

General strikes, by contrast, are inherently national by their organizational logic. They are

typically called by national labor federations or peak union bodies in response to national policy decisions such as austerity packages, labor law reform, pension cuts, or anti-privatization campaigns, and are designed to achieve simultaneous participation across all sectors and regions. The geographic structure of general strikes should therefore be strongly nationally homogeneous, not because spatial transmission fails but because the organizing logic is explicitly national. Political strikes targeting the government rather than a specific employer follow a similar logic and should similarly display national rather than geographic structure.

Public-sector and private-sector strikes represent a related dimension. Public-sector workers in many countries face a single national employer, are subject to nationally-set wage scales and budget cycles, and are organized through federations that negotiate at the national level. A government announcement of public-sector pay restraint creates the same mobilization moment simultaneously across states and provinces in a way that a private-sector wage dispute in one factory does not. Private-sector strikes, where wages are negotiated at the plant or sector level, should show more spatial heterogeneity.

Testing these distinctions is within reach using the existing LLM classification infrastructure. The URL and headline classification pipeline developed for event validation, which already identifies whether a GDELT event represents a genuine strike, can be extended to classify strike type using the same few-shot prompting approach applied to the same underlying article text and metadata. Classifying events as general or sectoral, public- or private-sector, industrial or political requires additional annotation effort and prompt engineering but no new architectural decisions. Industry classification from article text, covering sectors such as mining, transport, education, healthcare, and manufacturing, would permit a more granular decomposition.

Conclusion

This study has examined the question of whether strikes spread through geographic proximity or through broader national waves of mobilization. Descriptive patterns strongly suggest contagion. Regions with a striking neighbor are much more likely to strike than unexposed regions, and strike activity clusters visibly across space and time. Yet the central result of the analysis is that this apparent geographic diffusion does not survive a design that absorbs common national shocks. Once country-by-week fixed effects are introduced, the geographic neighbor effect disappears. The raw clustering of strikes is real, but it does not appear to reflect a process in which

mobilization travels step by step across adjacent regions. Instead, the evidence indicates that strike contagion operates primarily at the national level.

This finding contributes to the study of labor politics in two ways. First, it provides a systematic cross-national test of a claim that has been prominent in the comparative-historical literature but rarely evaluated with subnational causal designs. Labor historians and scholars of strike waves have long argued that major episodes of strike activity are nationally synchronized, shaped by common political opportunities, macroeconomic pressures, and organizational decisions that affect multiple regions at once. The results here support that view. When strike activity rises elsewhere in the same country, the probability of strike occurrence in a focal region rises substantially. By contrast, geographic adjacency does not add independent explanatory power once those national dynamics are taken into account. The implication is that the spatial clustering of strikes should not be interpreted, by default, as evidence of local diffusion. In the case of labor unrest, geographic proximity is often a marker of shared exposure to national forces rather than a conduit of transmission in its own right.

Second, the analysis helps clarify how strike diffusion differs from other forms of contentious politics. Research on civil conflict and some forms of protest has identified robust neighborhood effects that survive extensive controls, often because the underlying mechanisms are genuinely local: armed groups cross borders, refugees carry instability, ethnic ties span administrative boundaries, and violence diffuses through contiguous territories. Strikes are organized through different infrastructures. Trade unions, bargaining institutions, state labor policy, and national media systems are typically structured at the country level. That institutional architecture makes national synchronization more plausible than geographic spillover. The broader contribution of the paper is therefore not simply to document a null geographic effect, but to show that theories of contagion must be matched to the organizational and political structure of the phenomenon under study. A framework developed for civil war diffusion cannot be assumed to travel unchanged to labor mobilization.

The institutional moderation results extend this contribution by showing that institutions shape the spatial form of national strike waves even when they do not alter the strength of national contagion itself. Freedom of expression weakens the geographic concentration of strike activity, while freedom of association strengthens it. These patterns suggest that national waves can be routed through different social infrastructures. Where information circulates broadly through national media, strike signals reach workers across regions more uniformly. Where associational life

is stronger and more locally embedded, the same national mobilization may acquire a more geographically structured form. The contribution here is to move beyond the question of whether strikes diffuse and toward a more precise question about how nationally driven mobilization is distributed across space.

These findings point to a broader research agenda. If strike contagion is primarily national, then the next task is to explain the sources of national synchronization such as the economic shocks, political conflicts, organizational decisions, and bargaining cycles that generate country-wide waves of labor unrest. A second task is to identify the specific mechanisms through which media systems, union organization, and other institutional arrangements shape the geography of those waves. A third may be to disaggregate strike events by type. General strikes, political strikes, and sector-specific industrial disputes likely follow different diffusion logics, and aggregate analyses may conceal important heterogeneity. The main contribution of this study is to establish that the geography of strikes cannot be understood apart from the national structures in which they are embedded. That conclusion shifts the study of strike diffusion away from local proximity alone and toward the national institutions and mobilizing processes that organize collective action across space.

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Appendix

Summary Statistics

Table A1: Summary statistics for the analysis sample

	N	Mean	SD	Min	Max
Strike onset	315808	0.016	0.127	0.000	1.000
Neighbor strike (t-1, t-2)	315808	0.079	0.269	0.000	1.000
Freedom of expression (z)	294224	0.012	1.006	-2.434	1.013
Freedom of association (z)	294224	0.018	0.998	-2.537	0.928
State repression (z)	294224	-0.042	0.998	-2.372	1.103

Alternative Specifications

Table A2 replicates the baseline under two alternative specifications: two-way clustering of standard errors at both the ADM1 and country-week levels, and logistic regression. Results are qualitatively identical across all three. The null geographic neighbor effect is not an artifact of the standard error choice or the linear probability model.

Table A2: Robustness: alternative standard errors and estimator

Model:	strike_onset		
	(i)	(ii)	(iii)
	OLS	OLS	Logit
<i>Variables</i>			
Neighbor strike (t-1, t-2)	-0.0031	-0.0031	-0.0576
	(0.0021)	(0.0020)	(0.0660)
<i>Fixed-effects</i>			
ADM1	Yes	Yes	Yes
Country × Week	Yes	Yes	Yes
<i>Fit statistics</i>			
Observations	315,808	315,808	58,839
R ²	0.19665	0.19665	
<i>Signif. Codes: ***: 0.01, **: 0.05, *: 0.1</i>			

Validated-Location Robustness

The following tables replicate the main results restricting to strike events where the ADM1 geocode was confirmed correct by the LLM-based geo-validation step (`geo_quality == "validated"`). Unvalidated strike weeks are recoded as non-occurrence weeks, keeping the balanced panel intact. The validated sample has an occurrence rate of approximately 0.87% compared to 1.4% in the full sample. **Table A3** compares the baseline model across samples. **Table A4** compares the decomposition.

Table A3: Validated-location robustness: baseline specification

	strike_onset	strike_onset_val
Model:	(i)	(ii)
<i>Variables</i>		
Neighbor strike (t-1, t-2)	-0.0031	-0.0017
	(0.0021)	(0.0015)
<i>Fixed-effects</i>		
ADM1	Yes	Yes
Country $\times$ Week	Yes	Yes
<i>Fit statistics</i>		
Observations	315,808	315,808
R ²	0.19665	0.14678
<i>Clustered (ADM1) standard-errors in parentheses</i>		
<i>Signif. Codes: ***: 0.01, **: 0.05, *: 0.1</i>		

Table A4: Validated-location robustness: national vs. neighborhood decomposition

	strike_onset	strike_onset_val
Model:	(i)	(ii)
<i>Variables</i>		
Neighbor strike (t-1, t-2)	-0.0043** (0.0020)	-0.0011 (0.0014)
Country share (t-1, t-2)	0.6653*** (0.0507)	0.3052*** (0.0266)
<i>Fixed-effects</i>		
ADM1	Yes	Yes
Week	Yes	Yes
<i>Fit statistics</i>		
Observations	315,808	315,808
R ²	0.13931	0.07802
<i>Clustered (ADM1) standard-errors in parentheses</i>		
<i>Signif. Codes: ***: 0.01, **: 0.05, *: 0.1</i>		

Cross-Border Diffusion

Table A5 tests whether strikes spread across international borders. The sample is restricted to ADM1 units with at least one cross-border neighbor (~95 units). Columns (1)–(2) include only the cross-border exposure variable; columns (3)–(4) add within-country neighbor exposure jointly. Cross-border effects are null across all specifications.

Table A5: Cross-border diffusion

Model:	strike_onset			
	(i)	(ii)	(iii)	(iv)
<i>Variables</i>				
Cross-border neighbor strike (t–1, t–2)	0.0002 (0.0051)	0.0042 (0.0042)	0.0002 (0.0051)	0.0041 (0.0042)
Neighbor strike (t–1, t–2)			-0.0086 (0.0071)	0.0133*** (0.0039)
<i>Fixed-effects</i>				
ADM1	Yes	Yes	Yes	Yes
Country × Week	Yes		Yes	
Week		Yes		Yes
<i>Fit statistics</i>				
Observations	52,820	60,604	52,820	60,604
R ²	0.29890	0.11747	0.29904	0.11801
<i>Clustered (ADM1) standard-errors in parentheses</i>				
<i>Signif. Codes: ***: 0.01, **: 0.05, *: 0.1</i>				

LLM Classifier Prompts

Stage 1: URL Classifier (GPT-4o mini)

The URL classifier runs in two passes. Pass 1 uses the source URL alone. Pass 2 re-submits uncertain cases with the article headline added when available.

Pass 1 system prompt

You are classifying news article URLs to determine whether the article is about a labor strike, work stoppage, general strike, or boycott.

Relevant articles include: strikes that are ongoing, strikes that have ended or been resolved, strike aftermath, court orders or legal responses during a strike, and recent retrospectives on strikes from the past few years. All of these confirm a strike occurred in the contemporary period.

Classify each URL as exactly one of:

- relevant - the URL indicates the article covers a labor strike or work stoppage (ongoing, recently resolved, or recent retrospective)
- not_relevant - the URL clearly indicates an unrelated topic, OR the strike was explicitly called off BEFORE it started, OR workers explicitly stated they would NOT strike
- uncertain - the URL is too opaque to determine relevance, OR the strike is described as planned/threatened/balloted but it is unclear whether it actually happened

Rules:

- Base your judgment ONLY on the URL string - do not attempt to visit it.
- Recognize strike-related terms in any language (e.g., huelga, grève, greve, Streik, grev, sciopero, ,).
- Military, missile, or lightning strikes → not_relevant.
- A strike that ENDED, was SETTLED, or led to a CONTRACT AGREEMENT still occurred → relevant. Only classify as not_relevant if the strike was called

off or averted BEFORE it began.

- A strike being SUSPENDED mid-course (after starting) → still relevant.
- Court orders, injunctions, or bans issued DURING a strike → relevant.
- Articles explaining WHY workers are striking → relevant.
- Academic or purely historical analyses of strikes from the distant past (e.g., a 2025 article about a 1934 strike) → not_relevant.
- Strike PLANNED, THREATENED, or BALLOTTED but not confirmed to have started → uncertain.
- Hunger strikes, sit-ins, work-to-rule actions, and protest marches are relevant ONLY if the participants include workers or unions. The same action taken solely by non-worker actors (e.g., climate activists, political prisoners, students) without the participation of workers or unions → not_relevant.
- Union/labor mentions with no specific strike event → uncertain.
- Policy advocacy about strikes in general → not_relevant.
- Service disruptions without explicit strike mention → uncertain.
- Opaque URLs (numeric IDs, query strings with no descriptive slug) → always uncertain, never not_relevant.
- When genuinely in doubt, use uncertain.
- Respond with valid JSON only:

```
{"classification": "relevant"|"not_relevant"|"uncertain", "reasoning": "<one sentence>"}
```

Pass 1 few-shot examples *Relevant*

- chicagotribune.com: “chicago-teachers-strike-enters-third-day” — URL slug explicitly mentions a teachers’ strike.
- birgun.net: “grev-yasagi / emek-mucadelesi” — Turkish URL contains ‘grev-yasagi’ (strike ban) and ‘emek-mucadelesi’ (labor struggle).
- lemonde.fr: “greve-du-7-mars-sncf-ratp-enseignants” — French URL contains ‘greve’ and references transport and education workers.
- reviewjournal.com: “why-public-employee-strikes-should-be-illegal” — Opinion piece substantively about strikes.
- thehindu.com: “bank-employees-go-on-two-day-strike-over-wage-revision” — Slug explic-

itly names a bank strike.

- boston.com: “striking-boston-hotel-workers-reach-tentative-contract-agreement” — Workers were striking and reached an agreement; the strike occurred.
- bsccomment.com: “why-are-junior-doctors-striking” — Slug confirms doctors are actively striking.

Not relevant

- ghanaweb.com: “Dangote-s-refinery-to-begin-direct-petrol-supply” — Oil refinery supply logistics; no strike content.
- palsolidarity.org: “the-war-on-the-west-bank” — Israeli-Palestinian conflict; unrelated to labor action.
- bbc.com/sport/football/67890123 — BBC Sport football article.
- reuters.com: “us-inflation-hits-40-year-high” — Economic news; no strike indicated.
- oann.com: “ila-strike-suspended-dockworkers-agree-to-delay-walkout” — Walkout delayed before it began; no strike occurred.
- business-standard.com: “will-never-abstain-from-work-maha-lawyers” — Workers explicitly stated they would not strike.
- dissidentvoice.org: “the-1934-civil-works-administration-strike-in-utica” — 2025 article about a 1934 strike; historical analysis only.

Uncertain

- independent.ie: “asti-to-ballot-for-strike-action” — Ballot planned; strike not confirmed.
- dailypost.ng: “nigerian-academic-technologists-to-embark-on-strike-nov-14” — Strike announced for a future date; occurrence not confirmed.
- finance.yahoo.com: “union-workers-hawaiis-largest-hotel” — Mentions union workers but no strike indicated.
- shorouknews.com/news/view.aspx?cdate=...&id=... — Only a date and opaque numeric ID; no descriptive slug.
- news.bbc.co.uk/2/hi/business/7654321.stm — Numeric ID only; content unknowable from URL.
- aljazeera.com/economy/2023/5/14/article — Generic economy path; no slug describing content.

Pass 2 system prompt Pass 2 re-submits uncertain cases with the article headline added when it can be retrieved.

You are classifying news articles to determine whether they cover a labor strike, work stoppage, general strike, or boycott.

You will receive a URL and, when available, the article's page title.

Prioritize the title when present - it is more reliable than the URL slug.

[Classification categories and rules identical to Pass 1, with the following additions:]

- Strike PLANNED, THREATENED, or BALLOTTED but not confirmed to have started
→ uncertain. Only flip to relevant if the title explicitly confirms the strike began.
- Respond with valid JSON only:

```
{"classification": "relevant"|"not_relevant"|"uncertain", "reasoning": "<one sentence>"}
```

Pass 2 few-shot examples *Relevant*

- shorouknews.com (opaque URL), title: [Arabic: “Factory workers are on strike demanding higher wages”] — Arabic title states factory workers are striking for higher wages.
- news.bbc.co.uk (numeric ID), title: “UK bank workers to strike over pay” — Title explicitly confirms a strike.
- couriermail.com.au: “tram-services-wind-down-across-melbourne”, title: “Melbourne tram services grind to halt as workers strike” — Title confirms tram disruption is caused by a strike.

Not relevant

- allafrika.com (opaque URL), title: “Kenya: Government Announces New Infrastructure Budget” — Government budget; no labor strike.
- bellevuereporter.com: “boeing-renton-plant-to-halt-737-max-production”, title: “Boeing Renton plant to halt 737 MAX production” — Production halt due to manufacturing issues, not a labor strike.

Uncertain

- standaard.be (opaque ID), title: “Het Laatste Nieuws | De Standaard” — Title is just the newspaper homepage; no article content.
- vg-news.ru/n/141450, title: not available — URL is opaque and no title could be retrieved.

Stage 2: Geocode Validator (GPT-5 mini)

System prompt

You are a geolocation validator for GDELT strike event data. For each event you receive: a GDELT geocode (the location GDELT assigned to the event), the source URL, and any available article text. Your task is to determine whether the GDELT geocode correctly identifies where the strike occurred.

Output a JSON object with exactly these four fields:

```
"extracted_location" : where the strike actually occurred per the article
                        (city and/or region and country). Use "unknown" if
                        you cannot determine this.

"match"               : "yes", "no", or "uncertain"

"corrected_location" : when match is "no", the correct location in the
                        format "City, Region, Country" or "Region, Country".
                        Use null when match is "yes" or "uncertain".

"reasoning"          : one sentence explaining your judgment.
```

--- RULES ---

yes - The article confirms the strike occurred in the geocoded location or immediately adjacent area.

no - The article clearly states the strike occurred in a different location than the GDELT geocode.

uncertain - Use when:

- The article is inaccessible (broken link, paywall) AND the URL slug does not unambiguously name a specific location

- The strike is described as nationwide or multi-city and no single region can be confirmed
- The article is accessible but does not mention a specific enough location to confirm or contradict the geocode

Important:

- You can read and extract locations from articles in any language.
Non-English text is not a reason to return "uncertain".
- Do not use the news outlet's home country or city as a location signal.
- Do use the article dateline as a reliable location signal.
- GDELT frequently assigns events to capital cities when the actual location is elsewhere. Be alert to this pattern.
- For nationwide or general strikes, return "uncertain" rather than "no", even if GDELT assigned the capital.
- If the article is inaccessible but the URL slug or title unambiguously names a specific city or region, use that to determine the match.
 - Slug/title matches geocode → "yes"
 - Slug/title names a clearly different location → "no" with `corrected_location` set accordingly
 - Return "uncertain" only when the slug names just a country or is otherwise too vague.

Few-shot examples *Match: yes*

- GDELT: Sydney, NSW, Australia. Body text confirms strike at Westmead Hospital, Sydney.
- GDELT: Sudbury, Ontario, Canada. URL slug “sudbury-miners-go-on-strike-after-rejecting-deal-from-vale-canada” (inaccessible) unambiguously names Sudbury.
- GDELT: Colombo, Western, Sri Lanka. Article dateline reads “COLOMBO, Sri Lanka — Postal workers staged a protest...”
- GDELT: Paris, Île-de-France, France. French article describes SNCF workers striking in “région parisienne.”
- GDELT: Buenos Aires, Argentina. Spanish article: teachers striking in “Ciudad de Buenos Aires.”

- GDELT: Amman, Jordan. Arabic article: teachers striking in Amman.
- GDELT: São Paulo, Brazil. Portuguese article: metalworkers striking in São Paulo metropolitan area.

Match: no

- GDELT: Kabul, Afghanistan. Title: “Samangan Doctors on Strike, Demand Pending Salaries.” Corrected: Samangan, Afghanistan. Title names Samangan; GDELT assigned Kabul (capital city bias).
- GDELT: University of Botswana, South East. Article describes a strike at BIUST in Palapye, Central District. Corrected: Palapye, Central District, Botswana. GDELT confused BIUST with the University of Botswana in Gaborone.
- GDELT: Saint-Denis, Reunion. French article describes a garbage collectors’ strike in Marseille. Corrected: Marseille, Provence-Alpes-Côte d’Azur, France. GDELT assigned the outlet’s home location rather than the strike site.

Match: uncertain

- GDELT: Luanda, Angola. URL slug “angolan-teachers-begin-threeday-strike-over-wages” (inaccessible) identifies only the country; cannot confirm or contradict Luanda.
- GDELT: Naypyidaw, Myanmar. Article describes a nationwide general strike across Yangon, Mandalay, and dozens of cities; assigning to a single region is not meaningful.
- GDELT: Karingal, Victoria, Australia. Paywalled snippet mentions aged care staff at Baptcare striking; Baptcare operates across multiple states and the snippet is insufficient to confirm location.